

BIOMAP WORKSHOP · ICPR 2026 · LYON
BIO-INSPIRED METHODS FOR PATTERN RECOGNITION

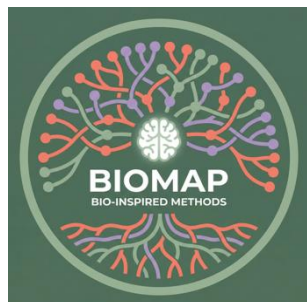
Breaking the Central Bias: Spatially Partitioned Experts for Coordinate-Based Neuroevolution

From a 4% central cluster to 72% image coverage with evolved spatial experts

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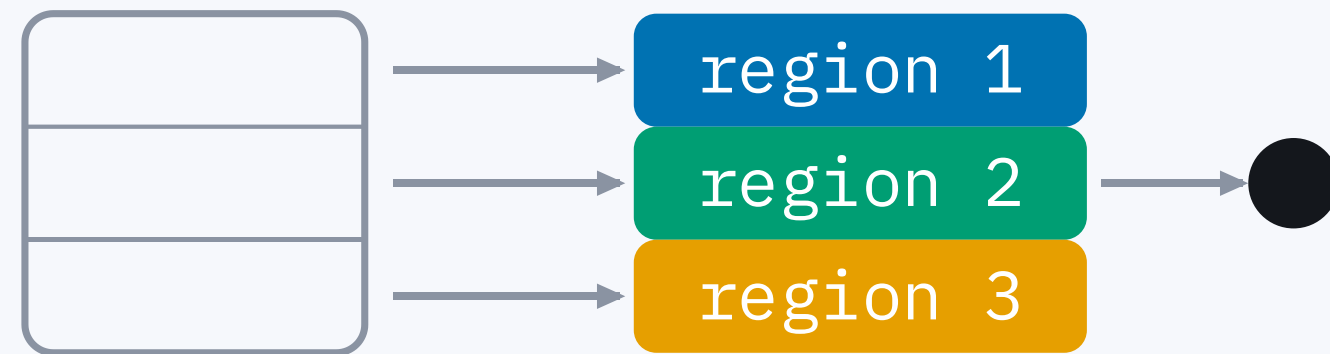
¹University of Neuchâtel, Switzerland · ²University College Dublin, Ireland

github.com/RomainClaret/es-hyperneat-optimization-studies



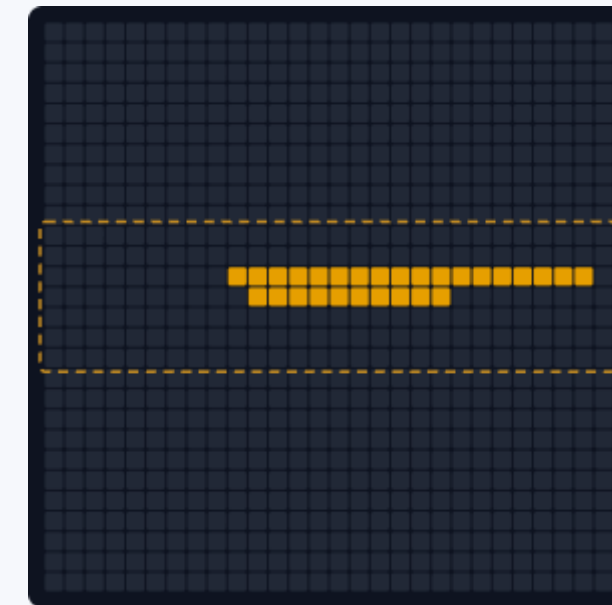
Brains don't process the retina monolithically. Evolved networks try to.

Biological vision



The visual field is **divided across specialized regions** ; perception is their composition.

A monolithic evolved network



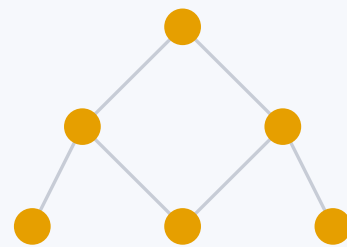
Given all 784 pixels, it converges on a **small central sliver** and ignores the rest.

Spatial-concentration bias : the network never uses features outside one small region. Accuracy alone won't tell you it happened.

Networks written as a function of geometry

STEP 1 · EVOLVE

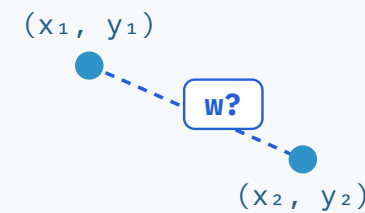
NEAT breeds a CPPN



A small **generator network**, bred by a genetic algorithm: mutation, crossover, selection. No backprop anywhere.

STEP 2 · QUERY

The CPPN answers geometry

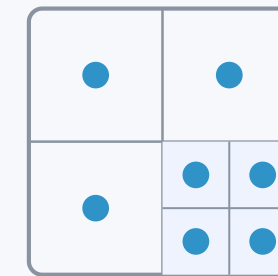


$$\text{CPPN}(x_1, y_1, x_2, y_2) \rightarrow \text{weight}$$

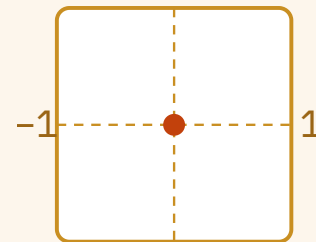
Ask it about any two coordinates and it returns the connection weight. **The network is a drawing over space.**

STEP 3 · GROW

A quadtree places neurons



Subdivide where the CPPN's output **varies**; place neurons only where patterns are non-uniform. Topology is discovered, never designed.



The detail that matters today: coordinates are normalized to $[-1, 1]^2$ with the origin at the **image center**. Every geometric computation radiates from the middle of the picture.

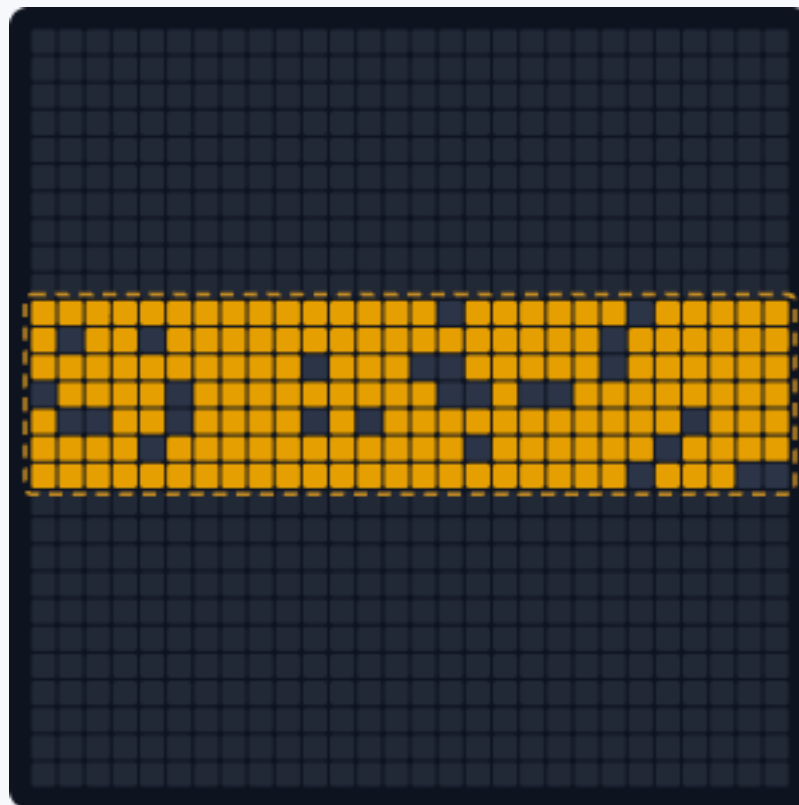
indirect encoding · no backprop

weights + topology both evolved

PUREPLES · pop 100 · 20 generations

21% accuracy, and every run stares at the same central band

UNION OF ALL 30 RUNS



169 / 784 pixels (21.6%), every one inside **rows 10–16** . The rest is invisible to the search.

20.95%

mean accuracy · 30 runs · max 29% · a linear classifier clears 90%

~48 px

used by the best networks, out of 784 available

More search doesn't help: early stopping makes runs **cheaper** , the ceiling **stays** . MNIST here is a microscope, not a leaderboard.

TWO POSSIBLE EXPLANATIONS

H1 · evolution is right

Ignoring peripheral pixels is an optimal strategy; central pixels suffice.

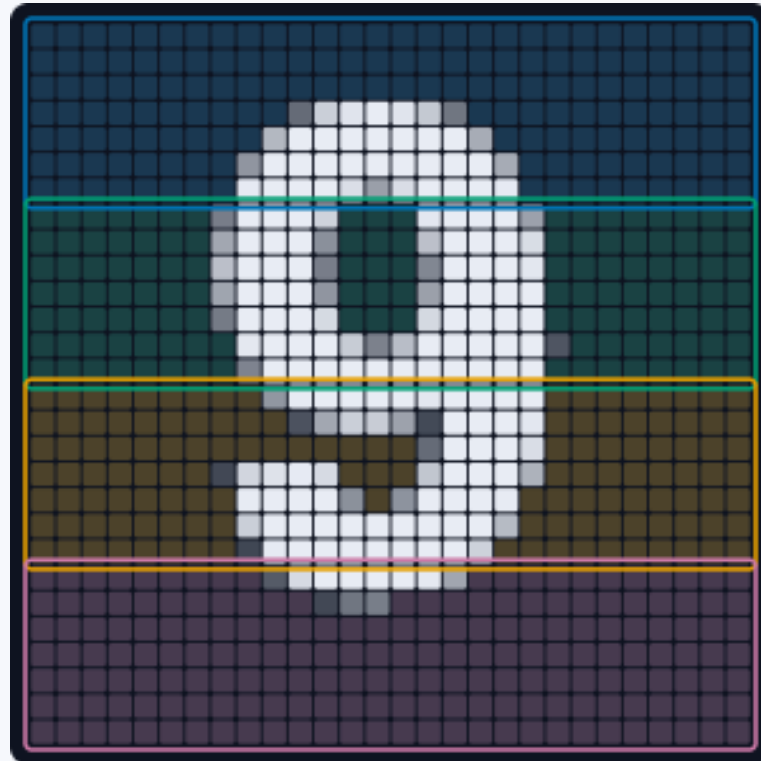
H2 · the architecture is the ceiling

A monolithic network cannot integrate information across a wide receptive field.

→ we test H2 with a divide-and-conquer intervention

Cut the input into bands; give each band its own expert

THE 28×28 IMAGE · $N_e = 4$ SHOWN



one color per expert band

FIXED ASSIGNMENT · BAND $k \rightarrow$ EXPERT k



Deterministic spatial routing , no learned gate. MoE-inspired, a spatially partitioned ensemble.

1D slicing is **deliberately minimal** : it isolates “any partitioning at all” from geometry-aware designs.

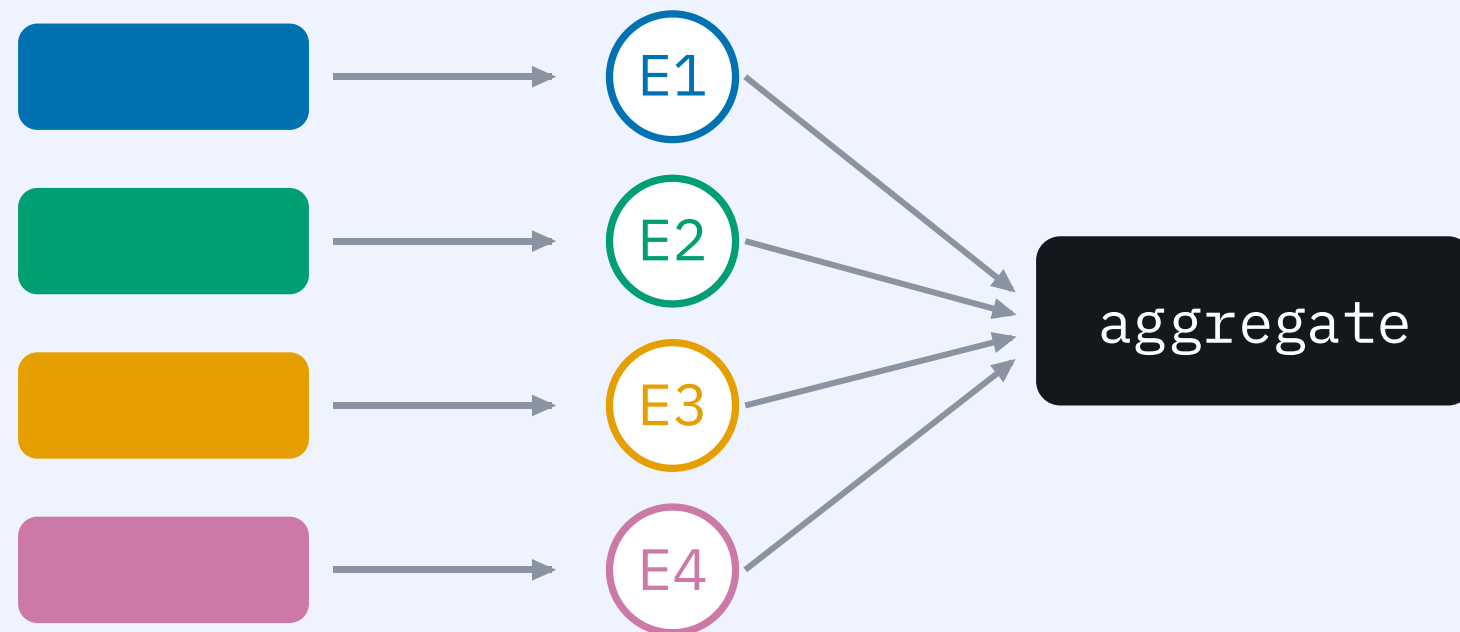
N_e swept 1 → 17

$N_e=1 \Leftrightarrow$ baseline replica

$N_e=17 \rightarrow \sim 46$ px/segment

Same partitions, two architectures

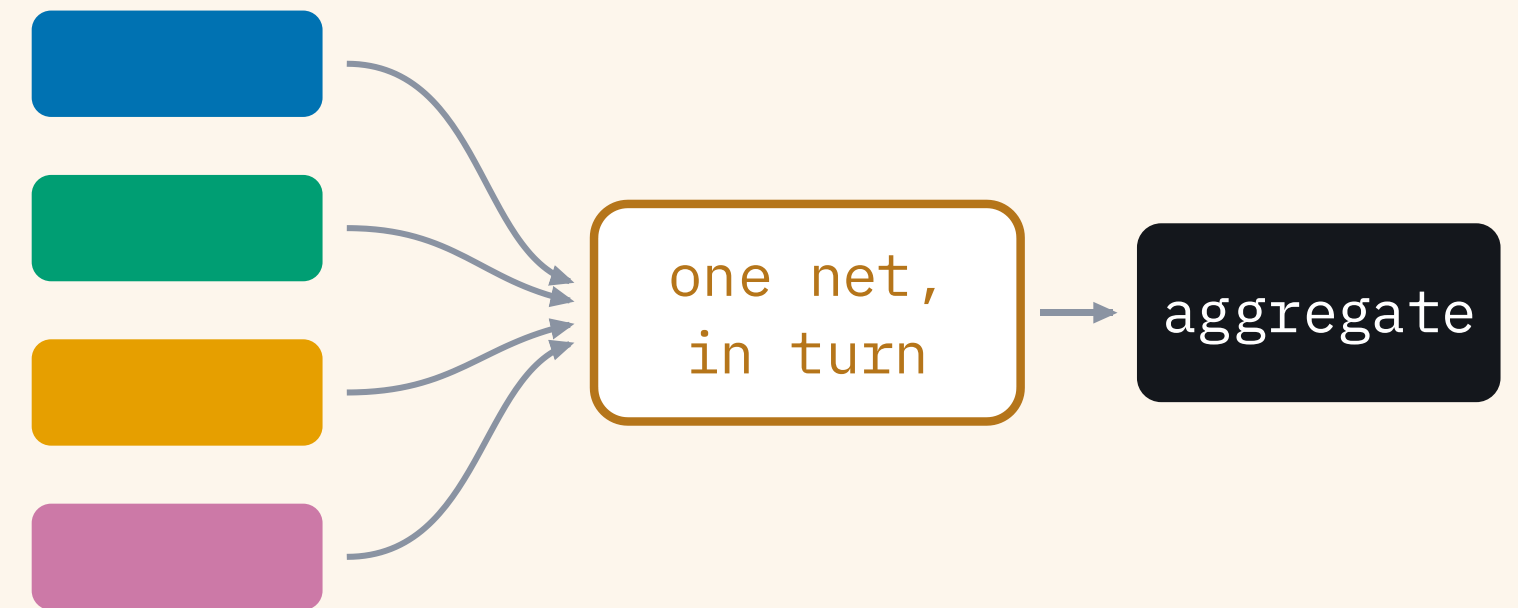
Independent MoE



one evolving population per segment

One **separately evolved specialist per segment** : parallel populations, cooperative-coevolution style.

Shared MoE



⌚ no partition-identity input

A **single evolved network** processes every segment sequentially. Same partitions, no specialization.

hyperparameters pinned to baseline · no re-tuning

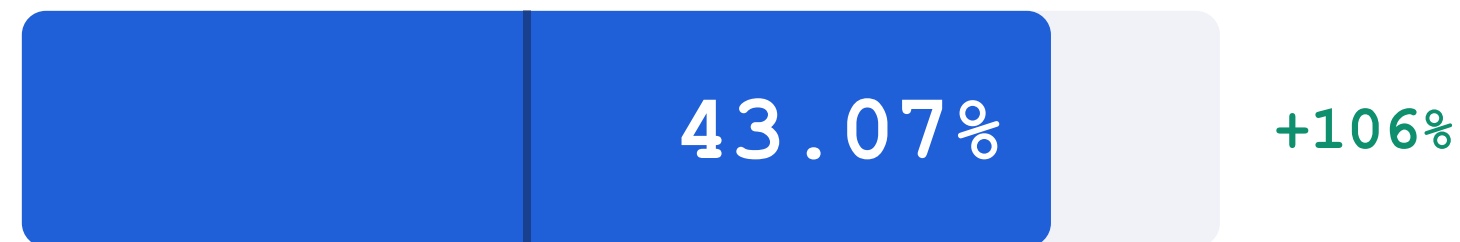
batch controls: SBG · SBE · SM → A5

30 replicates × 7 conditions + baseline

Specialization doubles accuracy. Partitioning alone doesn't.

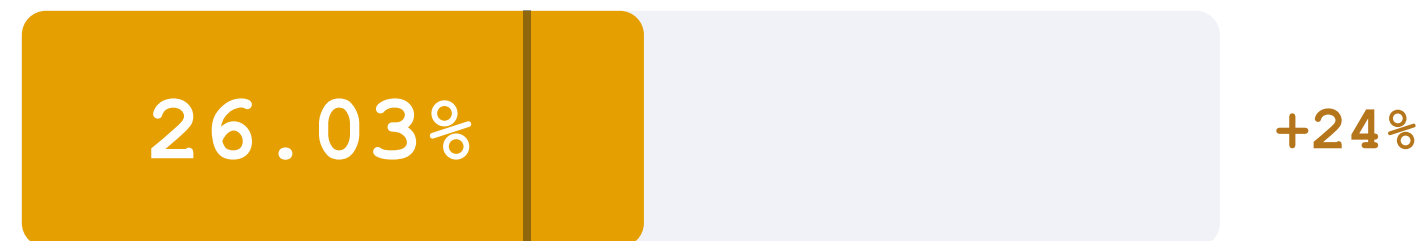
Independent MoE

13 experts · PW
aggregation



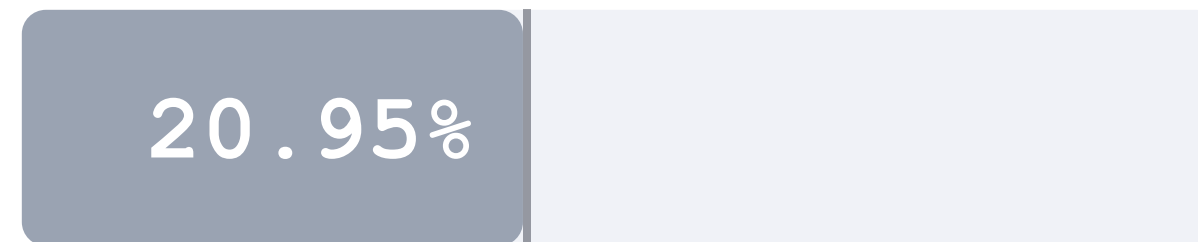
Shared MoE

one net, no identity



Monolithic baseline

prior work · 30 runs



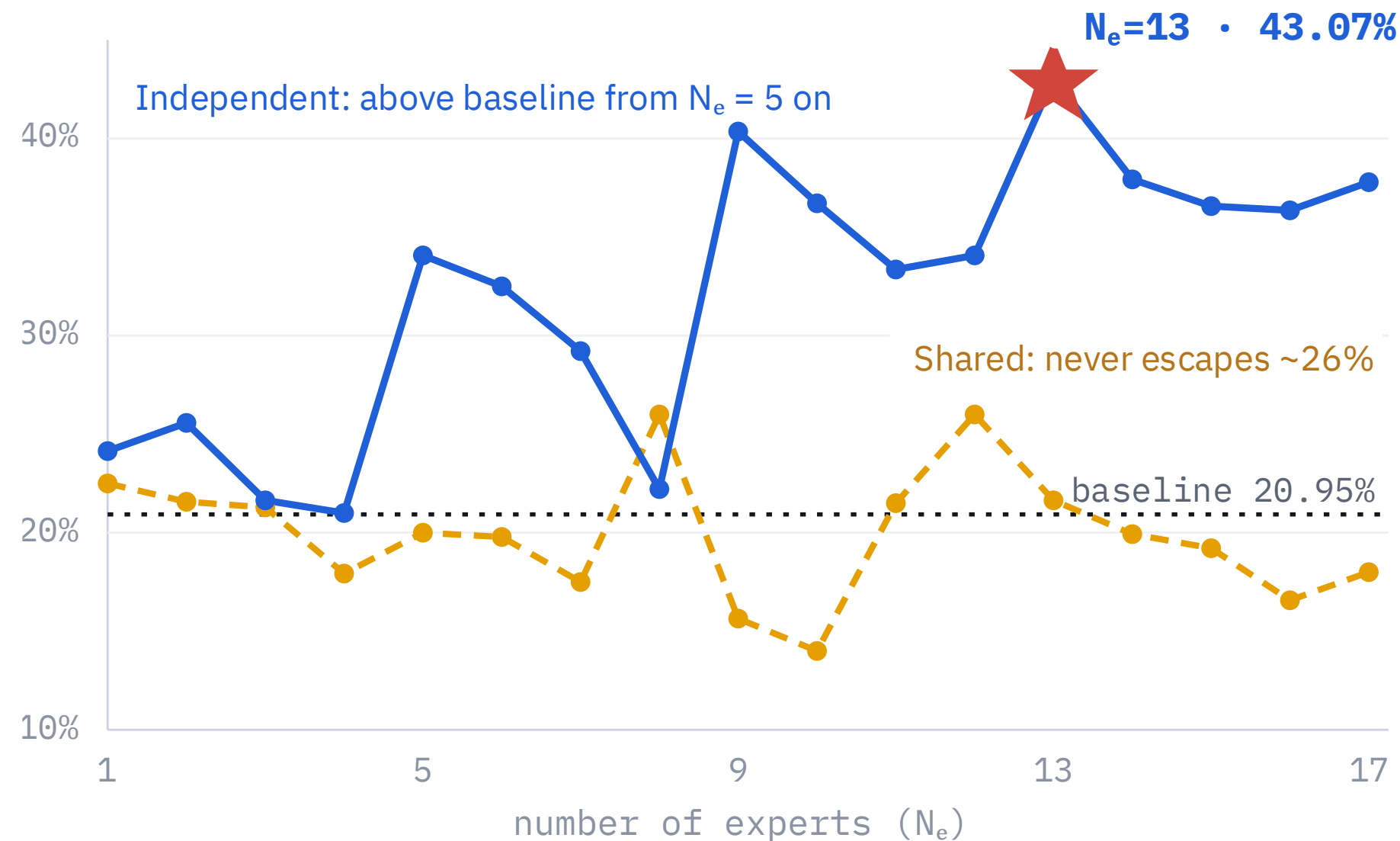
dashed marker = monolithic baseline, 20.95%

Same partitions, same budget, same hyperparameters. **The gain comes from independent specialists**, one per partition.

The **worst** Independent run (38.0%) beats the **best** baseline run ever recorded (29.0%).

full spread & tests → A1 · A2

A sharp peak at 13 experts, and a capacity story behind it



784 / 13 $\approx$ 60 px

per expert at the peak, the same order as the baseline's ~48-px converged field

A practical **search-capacity ceiling** : under this budget, evolution fully exploits ~50–60 input dimensions. Partition to match it.

Coarser $\rightarrow$ the bias re-emerges inside each segment.
Finer $\rightarrow$ too few pixels to tell a 1 from a 7. $\rightarrow A3$

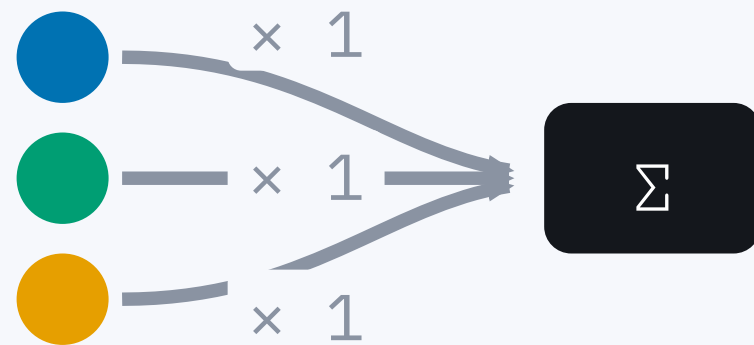
real data · best aggregation per N_e · Default conditions

Architecture creates the gain; aggregation amplifies it

FAMILY 1

Simple heuristics

Avg · Sum · Max



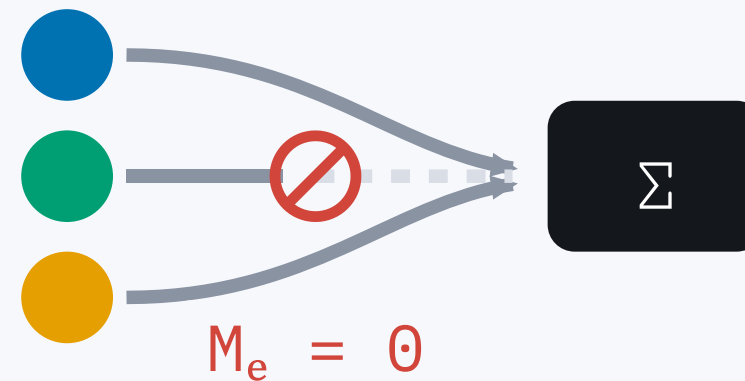
Every expert counts equally. No data needed, and erratic as the expert count grows.

35.60% *best Avg* · +70%

FAMILY 2

Mask-based

Conn-Masked · Conn-Weighted · Thresh-Masked



Silence experts with no structural connection to a class:

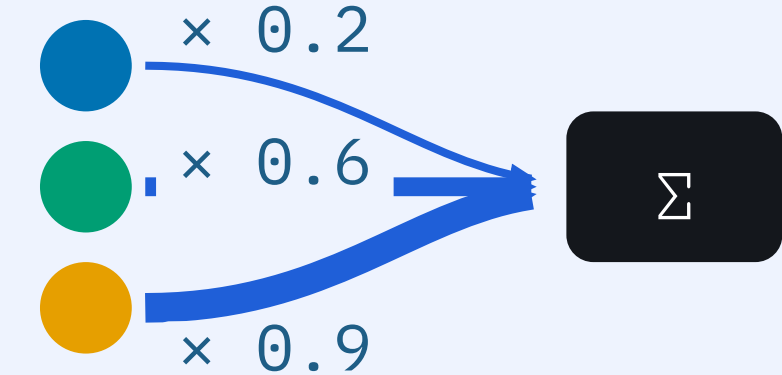
$$S_{\text{agg}} = \sum (S_e \odot M_e).$$

mid-pack

FAMILY 3 · WINNER

Performance-weighted

Perf-Weighted · Top-Expert · Evolved-Wt



Weight expert e on class c by its validation F1:

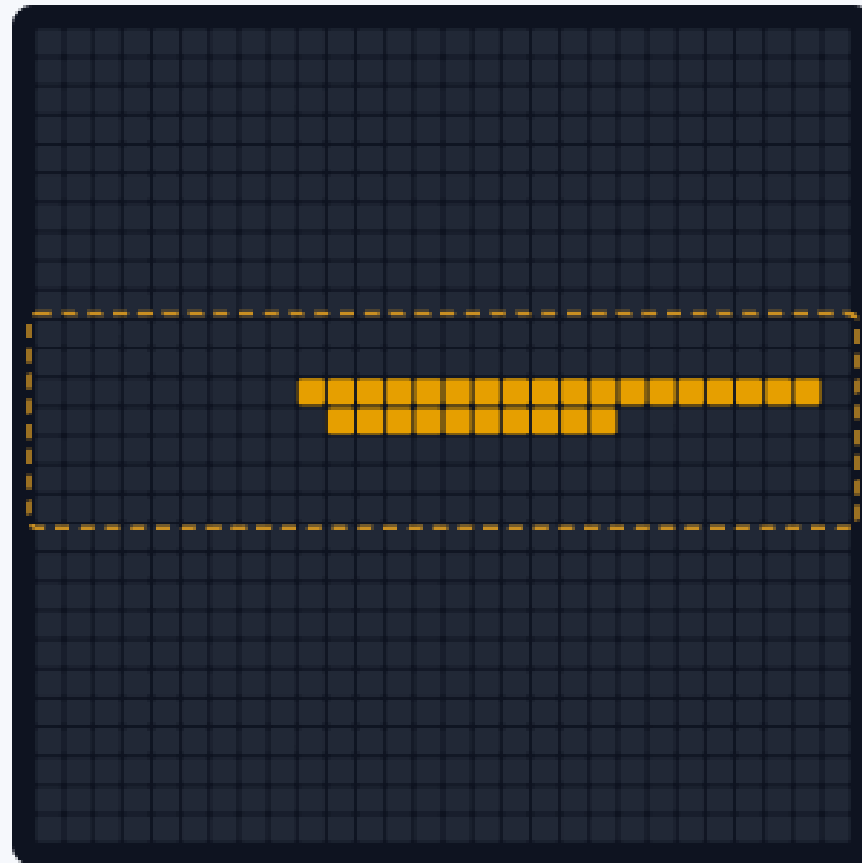
$$S_{\text{agg}}[c] = \sum W_{e,c} \cdot S_e[c].$$

43.07% vs 32.93% Avg, same nets

Equal-weight averaging uses **no validation data** and still beats the baseline by **+70%**. The partitioned architecture carries the gain on its own. full 14-method ranking → A4

Partitioning forces evolution to see the whole image

Monolithic baseline



4% 28 of 784 pixels one or two thin central lines

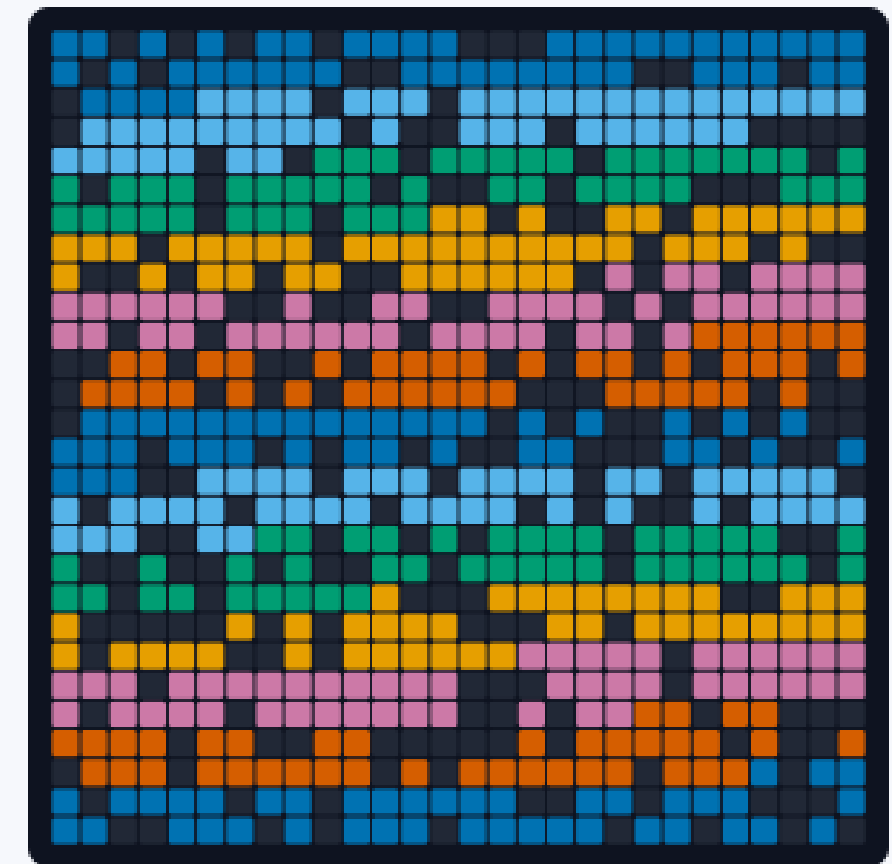
all 30 runs land in rows 10-16

13 spatial experts



one specialist per segment

Independent MoE · 13 experts



72% 562 of 784 pixels 10× the baseline · peak 79% at $N_e=11$



20.95% → **43.07%** mean accuracy · **+106%** · same encoding, same budget

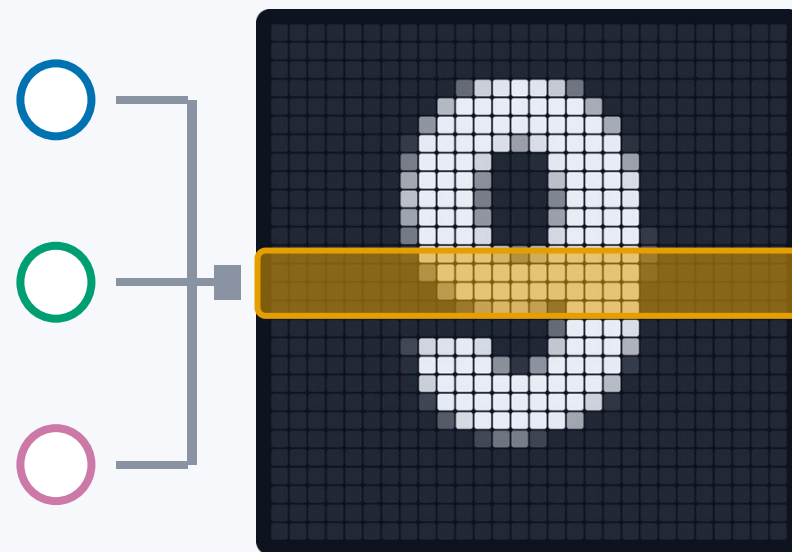
Forced spatial diversity, not ensemble size

ensemble error = average error - **diversity**

KROGH & VEDELSBY

Unconstrained experts

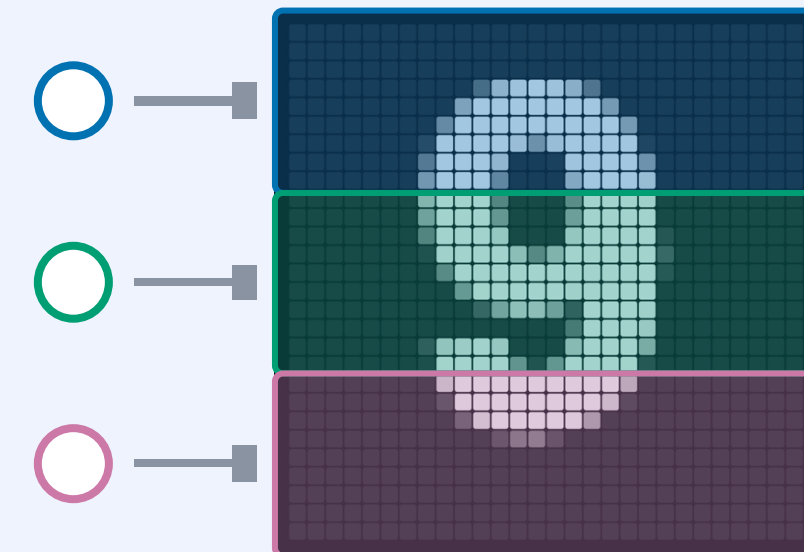
diversity ≈ 0



All 30 baseline runs converge on the **same central band** (union: 169/784 px). The ensemble reduces to one model wearing thirteen hats.

Partitioned specialists

forced diversity



Each expert can only mine **its own band** : complementary by construction. Individuals stay near baseline; the gain is the diversity. Shared MoE agrees: 26%, not 43%.

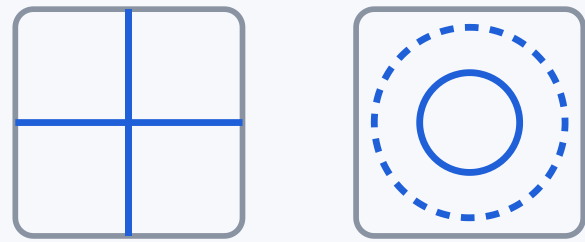
Central bias is an architectural limitation . The evolutionary search isn't the bottleneck.

honest limit: one dataset; MNIST centering
aligns both biases → off-center data next
alternative CPPN-regularity account → [A7](#)

Three direct extensions

01

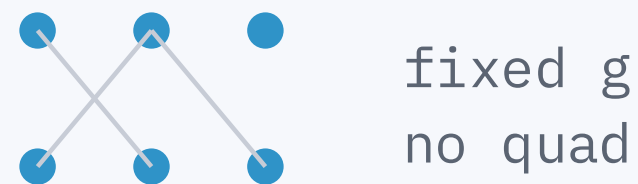
Geometry-aware partitions



Quadrant, block, radial, or **evolve the boundaries** jointly with the experts.
The 106% from blind 1D slicing is a **lower bound**.

02

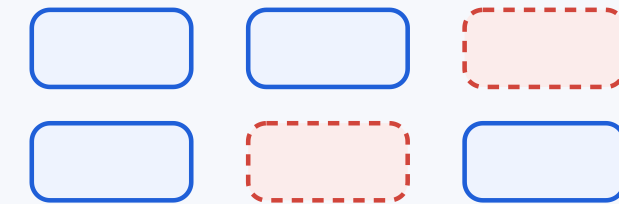
Plain-HyperNEAT discriminator



No variance-driven placement $\Rightarrow$ no central bias by that route. Fails at $\sim 21\% \Rightarrow$ CPPN smoothness is the ceiling; does better $\Rightarrow$ **the bias is ES-specific**.

03

Mechanism-isolating ablations



Non-symmetric activation palettes, variance-threshold annealing, center-shifted CPPN bias inputs. **Which ingredient makes the bias?**

Natural larger-scale target: **biomedical imagery**, where peripheral regions are clinically informative, exactly what a centrally biased model misses.

Diagnose the collapse. Partition to break it.

The diagnostic

Receptive-field tracking : extract each champion's active input pixels per generation. Exposes silent input-coverage collapse that accuracy alone hides.

a 10-line plot would have caught this on day one

The remedy

Spatial partitioning : windows sized to the search's capacity (~50–60 dims here), one evolved specialist per window, votes weighted by validation F1.

4% → 72% coverage · 20.95% → 43.07% accuracy

coverage **4%** → **72%**

accuracy **20.95%** → **43.07%**

sweet spot $N_e = 13 \approx$ search capacity

Merci. Questions?

Romain Claret · BIOMAP Workshop · ICPR
2026 · Lyon · 22 Aug

github.com/RomainClaret/es-hyperneat-optimization-studies

code · configurations · data · fully reproducible

All conditions, Perf-Weighted vs naive Avg

CONDITION	PW MEAN±SD	PW BEST/WORST	AVG MEAN±SD	AVG BEST/WORST
Independent · Default	43.07 ± 2.45	49.0 / 38.0	32.93 ± 3.29	39.0 / 27.0
Independent · SBG	41.23 ± 3.65	50.0 / 35.0	35.60 ± 3.31	42.0 / 29.0
Independent · SBE	41.90 ± 3.26	50.0 / 35.5	29.60 ± 2.24	33.5 / 24.5
Shared · Default	26.03 ± 2.06	30.5 / 22.0	22.52 ± 1.03	24.0 / 20.0
Shared · SBG	23.88 ± 1.50	27.0 / 21.0	23.15 ± 2.57	28.5 / 19.0
Shared · SM	16.75 ± 2.20	21.0 / 12.5	13.93 ± 2.64	19.0 / 9.0
Shared · SBG+SM	19.22 ± 2.50	25.5 / 15.0	13.27 ± 1.84	16.5 / 8.5
Monolithic baseline	20.95 ± 2.41	29.0 / 17.0	20.95 ± 2.41	29.0 / 17.0

all values in % · PW = Perf-Weighted (F1-based) · median±MAD in the paper, Table 1

The numbers behind the claims

FAITHFUL REPLICATION · $N_e=1$ vs original

$t = -0.68$ · $p = .502$ · $d = -0.17$

ARCHITECTURE EFFECT · two-way ANOVA

$F(1, 3536) = 11\,915.88$

ARCHITECTURE × GRANULARITY

$F(16, 3536) = 114.43$

EXPERT COUNT · one-way + Tukey HSD

$F(16, 493) = 230.78$ · $N_e=13$ superior to all

BEST MoE vs REPLICATED BASELINE

$t = 33.69$ · $d = 8.70$

BEST INDEPENDENT vs BEST SHARED

$t = 29.19$ · $d = 7.54$

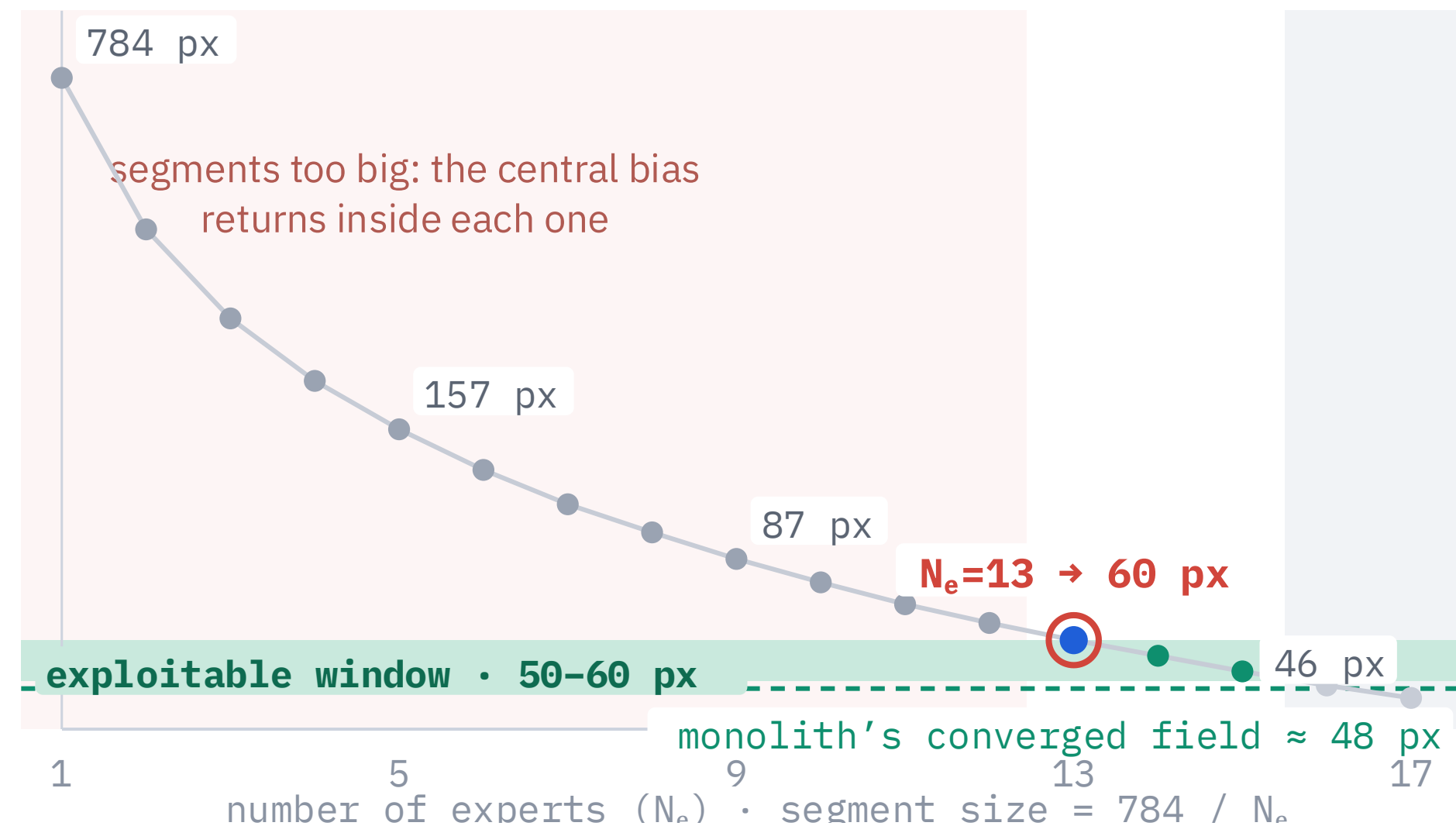
AGGREGATION EFFECT · 14 methods

$F(13, 406) = 798.06$ · PW vs Avg: $t = 13.54$,
 $d = 3.50$

CONVENTIONS

Welch's two-sided t · Cohen's d (0.2/0.5/0.8 = small/med/large) ·
sig. at $p \leq .05$ unless noted

Why the peak sits at 13 experts



Under this budget the search fully exploits **~50–60 input dimensions**. The baseline's ~48-px converged field marks the same limit from below.

$N_e = 13 \rightarrow 60 \text{ px}$: the **largest segment that still fits the window**. Coarser (gray): the bias returns inside each segment. Finer (16–17): too few pixels to tell a 1 from a 7.

Tukey HSD: $N_e = 13$ (43.07 ± 2.45) statistically superior to **every other granularity tested**.

Clear stratification: F(13, 406) = 798.06

mean accuracy at $N_e = 13$

F1-weighted · top tier

up to 43.07%

Perf-Weighted · 43.07%

Top-Expert

Evolved-Wt

Struct-Gated

Improve with granularity by exploiting specialization. All peak at $N_e = 10-13$.

Mask-based · mid-pack

structural gating

Conn-Masked · binary

Conn-Weighted · boosted

Thresh-Masked · strength

Silence experts with no structural route to a class: $S_{agg} = \sum (S_e \odot M_e)$.

Simple heuristics · erratic

no data needed

Avg · 32.93%

Sum

Max

Equal votes, conflict-blind as the expert count grows. Still +70% over baseline at best; 10 of 14 methods shown.

AT $N_e = 13$, SAME NETWORKS

43.07%

Perf-Weighted

32.93%

Avg · $t = 13.54$, $d = 3.50$

Weighting is per class: $W_{e,c}$ = expert e's validation F1 on class c. Structure-Gated additionally requires structural connectivity; Evolved-Wt adds a per-expert coefficient (hand-set here).

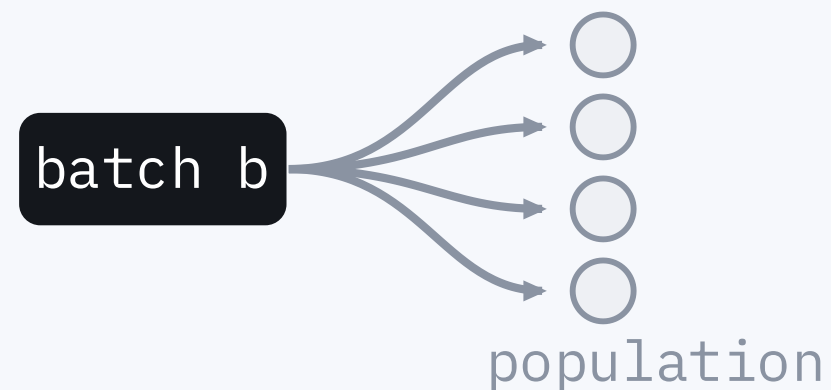
Honest gap: the gating is static by design. A learned gate might find better assignments. We traded that for a search space evolution can afford.

Isolating architecture from data-presentation variance

SBG

both architectures

Same-Batch-per-Generation

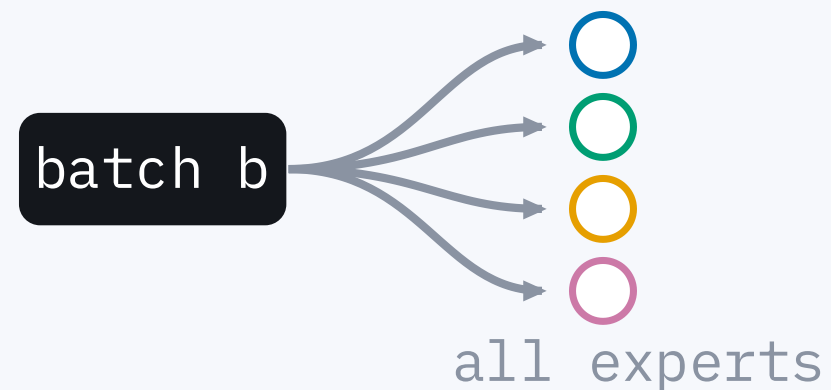


All individuals in a generation score on one once-drawn batch: a stable fitness landscape, no sampling variance between individuals.

SBE

Independent only

Same-Batch-per-Expert

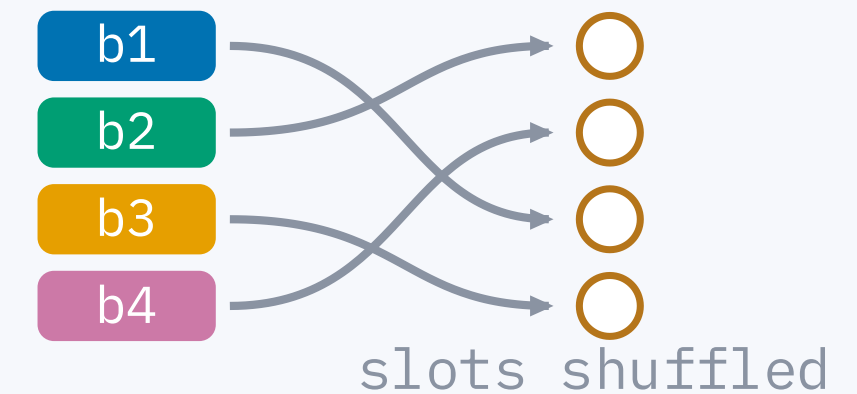


Every expert sees the same batch, a fair cross-expert comparison. Enabling SBE enforces SBG.

SM

Shared only

Shared-Mix



Shuffle batch indices per expert slot, breaking the fixed segment-to-slot pairing to prevent unwanted coadaptation.

Default (noisy batches) **beats SBG** for PW: 43.07 vs 41.23 ($t = 2.29$, $d = 0.59$). Fresh batches per individual select for generalization rather than memorization.

Default vs SBE: not significant ($p = .122$). Noise helps the population escape local optima; stabilized batches mainly help naive Avg (35.60).

Three properties that compound

i

Origin-anchored frame. Inputs are normalized to $[-1, 1]$ around an origin that coincides with the geometric middle of the image, the standard substrate convention.

ii

Symmetric primitives × variance-driven placement. Gaussian and sine CPPN activations produce structured spatial variance that the quadtree's subdivision responds to, following the primitive's symmetry.

iii

No pressure to leave. Once central pixels suffice (as on MNIST), no fitness signal pushes CPPN bias inputs to shift the response off-center, so the search settles instead of continuing.

Stated as a **hypothesis**. The ablations (slide 12) attribute it, and the falsification test is to swap in **asymmetric primitives** and ask whether the central basin survives. Partitioning fixes it one level up, whichever mechanism dominates.

Two counts, one search: 48 px (6.1%) counts the best networks' fields; 28 px (3.6%) is the stricter coverage convention used in our figures.

Could it just be CPPN regularity? Partly. Not solely.

The hypothesis (Clune et al.)

CPPN encodings degrade as target-pattern **regularity decreases** ; complex perceptual tasks sit in that regime. Partitioning would then help by shrinking each expert's substrate. Central bias would be a surface symptom.

THE CROSS-CHECK

	smaller substrate?	large gain?	mean
Independent	✓	✓	43.07%
Shared	✓	✗	26.03%

Same substrate reduction, different outcome: **smoothness can't be the sole driver.**

What survives of it

The regularity ceiling stays real: each per-expert CPPN inherits it, capping how far any partitioning can push.

caveat: Shared re-applies one CPPN, not a perfectly clean comparison

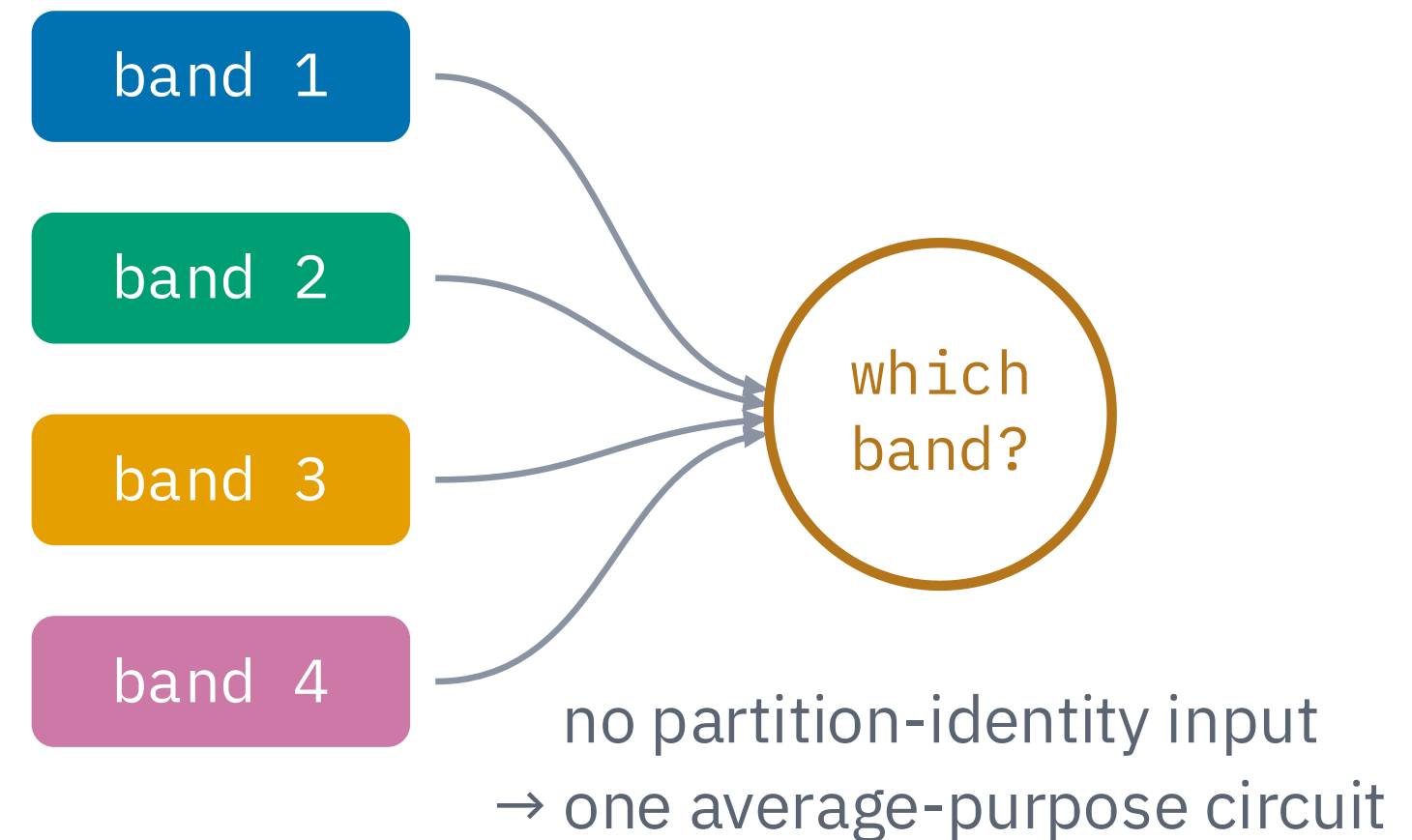
The cleanest discriminator between the two accounts: the **plain-HyperNEAT baseline** (slide 12): fixed substrate, no quadtree. Fails near 21% ⇒ smoothness is the ceiling; does better ⇒ the bias is ES-specific.

Partition identity is information the Shared net never gets

The shared network processes disjoint segments **sequentially, anonymously**. Weights must average over all bands instead of specializing.

Result: 26.03% best case, a +24% bump from partitioning alone, far from the +106% with specialists.

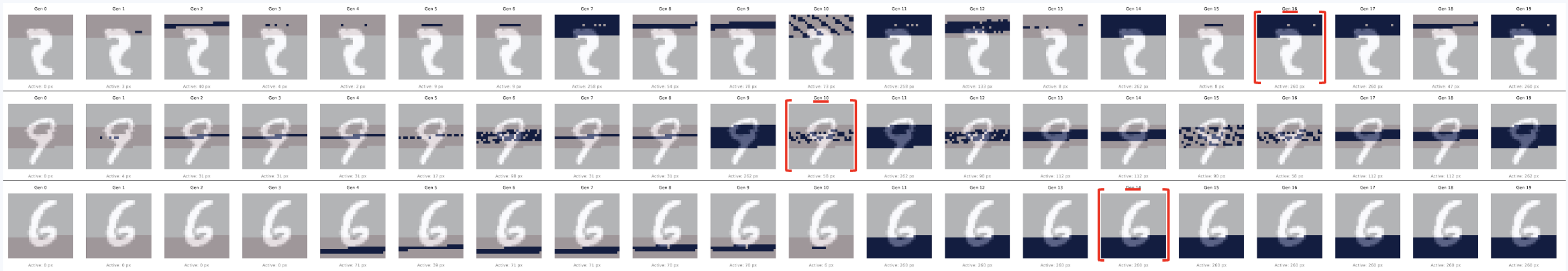
Shared-Mix (decorrelating slots) drops it to **16.75–19.22%**, below baseline. The fixed segment-to-slot pairing was doing hidden work.



Watching coverage grow, generation by generation

CHAMPION ACTIVE INPUTS · $N_e=3$ · ONE ROW PER EXPERT · GEN 0 → 19

red box = convergence



28 px

$N_e=1$ · 3.6% · the monolithic habit

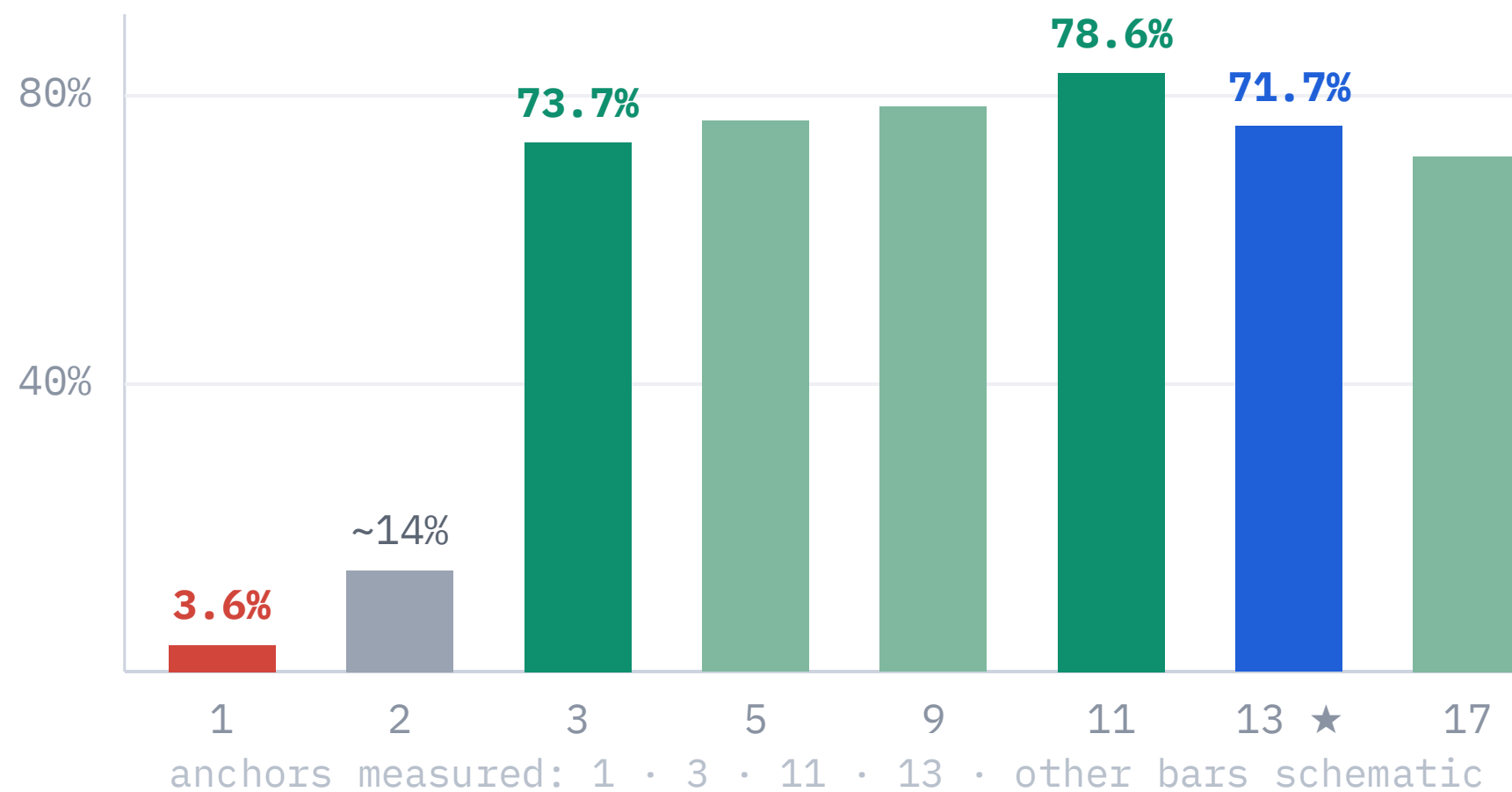
578 px

$N_e=3$ · 73.7% · crossover between 2 and 3 experts

79% / 72%

coverage peaks at $N_e=11$; the accuracy peak (13) trades coverage for refinement

Coverage growth is sharp, then non-monotonic



The crossover sits **between 2 and 3 experts**: at $N_e=2$ each half still exceeds the ~50–60-px capacity window, so the central habit persists.

Coverage peaks at $N_e=11$ (78.6%); accuracy peaks at 13 with less coverage (71.7%): **broad coverage trades against refined features** .

Composite at 13 experts: **562 px active** , 10× the baseline's 48, though each expert sees only ~60.

Pinned to the TPE-optimized baseline, deliberately

NEAT

```
population 100 initial hidden
nodes 0 conn add / delete 0.5 /
0.5 node add / delete 0.8 / 0.2
initial connections: direct, 80%
default activation tanh
```

ES-HyperNEAT substrate

```
initial depth 2 · max depth 5
variance threshold 0.01 division
threshold 0.5 max weight 3.0
iteration level 0 activation
tanh
```

TPE (Tree-structured Parzen Estimator) found this configuration in the baseline study. Bayesian and sample-efficient.

Reused **unchanged** everywhere. Optimized for the monolith, so any MoE gain is structural, not tuned.

```
PUREPLES (ES-HyperNEAT ref.
impl.) 20 generations · 30
replicates rest: NEAT-Python
XOR defaults
```

Input-side modularity for an adaptive substrate: the new bit

HyperNEAT lineage

Stanley '09: weights from geometry on a fixed substrate. Risi & Stanley '12 (ES): topology evolved too, via quadtree. Our substrate is theirs. The input is what we change.

Modularity in neuroevolution

Verbancsics '11: constrained internal connectivity. Reisinger '04: reusable modules. MM-NEAT '16: behavioral modules at the output. We apply modularity to the **input itself**.

Mixture-of-Experts

Jacobs '91 → sparsely-gated MoE '17: learned routing to specialists. Ours is **deterministic geometric routing**. In this setting diversity must be forced, or experts collapse together.

Cooperative coevolution

Potter & De Jong '00: decompose, evolve sub-populations, combine. Our decomposition is spatial, and post-evolution aggregation **sidesteps credit assignment**.

2002

NEAT

2009

HyperNEAT

2012

ES-HyperNEAT

2026

+ spatial experts (this work)

A diagnostic instrument, not a benchmark entry

Why it's the right instrument

Centered, size-normalized digits give a **clean, reproducible** instance of central collapse, plus prior baselines (20.95%) to replicate against. Cheap enough for 10 000+ evolutionary runs.

What it can't settle

Digit centering aligns the **coordinate-system bias** with the **information-density bias** ; they point the same way. Separating them needs datasets with off-center informative regions.

What we claim

Not absolute accuracy; a linear classifier clears 90%. The claim is **mechanistic** : the ceiling is architectural, partitioning removes it, and both tools transfer to other coordinate-based systems.

Context within neuroevolution (30-run means): HyperNEAT generic **23.9%** · HyperNEAT-LeNet **27.7%** , both hand-designed substrates. The partitioned **43.07%** clears both with a substrate that was **discovered**, using ~100× fewer neurons.